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School of Computer Science, Engineering and Applications (SCSEA)
MCA FY (AIDS)
Subject: Advance Big Data Analytics

Case Study :	**Detecting Fake Internship Postings Using NLP and Big Data Analytics**	
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1. Introduction

The rapid expansion of digital job portals and internship listing platforms has fundamentally transformed how students search for career opportunities. Platforms such as LinkedIn, Internshala, and Naukri make it easy to discover and apply for internships, but this convenience has also opened the door to fraudulent activities. Many malicious actors now post fake internship listings to exploit students by collecting personal information, charging fake application fees, or committing identity theft. Studies indicate that a significant percentage of job postings—sometimes over 17%—may show signs of being fraudulent, highlighting the seriousness and growing scale of this issue.

To address this problem, technologies like Natural Language Processing (NLP) and big data analytics provide an effective and scalable solution. NLP helps analyze the content of job postings to detect suspicious patterns such as vague descriptions, unrealistic salary offers, and urgency-driven language, while big data tools like Apache Kafka and Apache Spark enable real-time processing of large volumes of data. This case study proposes a fake internship detection system that combines machine learning, deep learning, and distributed data processing. Using the EMSCAD dataset, the system achieves an accuracy of 95.4%, demonstrating its effectiveness in identifying fraudulent listings and protecting students from potential scams.

2. Case Study Description

2.1 Problem Background

Students are increasingly targeted by fraudulent job offers delivered through popular internship and job listing platforms. Fake postings often impersonate well-known companies, using professional language and branding elements to appear legitimate. These listings frequently promise high stipends, immediate selection, and flexible remote work arrangements, creating an illusion of an exceptional opportunity to deceive applicants.

2.2 Characteristics of Fake Internship Postings

Research into fraudulent postings reveals several distinguishing characteristics:

- Unrealistic compensation promises (e.g., Rs. 50,000/month for freshers with no skills required)
- Vague or absent company descriptions with no verifiable website, address, or registration
- Requests for personal or financial information during the application process
- Poor grammar, inconsistent formatting, and excessive use of superlatives
- Urgent hiring language such as "Immediate joining" or "Limited seats available"
- Lack of specific qualifications or role requirements
- Use of generic email domains (Gmail, Yahoo) instead of official corporate emails

2.3 Scale and Impact

The sheer volume of internship listings posted daily makes manual verification impractical. Thousands of new postings appear every day across major job platforms. Students who fall victim to fraudulent listings risk financial loss, identity theft, and psychological distress. A scalable automated system is therefore essential to screen listings at the point of submission, before they reach vulnerable applicants.

2.4 Existing Limitations

Current fraud detection mechanisms typically rely on user reports and basic keyword filters. These reactive approaches allow fraudulent listings to remain active for days before detection. There is a clear need for a proactive, intelligent, data-driven system that classifies postings automatically in real time using semantic understanding of text.

3. Proposed Solution

The proposed solution is a multi-stage intelligent pipeline that ingests raw internship postings, processes them using NLP techniques, extracts meaningful features, and classifies each posting as genuine or fraudulent using an ensemble machine learning model.

3.1 Solution Architecture Overview

The architecture consists of five primary layers working in sequence:

- Data Collection Layer: Web scraping using Scrapy and BeautifulSoup to collect postings from multiple job portals, supplemented by a direct submission API for platforms adopting the service.
- Pre-Processing Layer: Cleaning raw text by removing HTML tags, special characters, and irrelevant tokens. Applying tokenization, stop word removal, and lemmatization using NLTK and spaCy.
- Feature Engineering Layer: Extracting numerical features from text using TF-IDF vectorization, Word2Vec embeddings, sentiment polarity scores, and Named Entity Recognition to detect company names, locations, and contact information.
- Big Data Processing Layer: Apache Kafka serves as the real-time message broker for ingesting new postings as streaming events; Apache Spark processes them in parallel across a distributed cluster.
- Classification Layer: An ensemble of Random Forest, SVM, and LSTM neural network models classifies postings, with a final majority-vote decision producing the output label and confidence score.

3.2 NLP Strategy

The NLP component forms the intellectual core of the system. Text fields including job title, description, requirements, and company profile are independently analyzed. Sentiment analysis identifies overly positive or urgency-based language. NER detects whether a real company name and physical address are present. TF-IDF highlights statistically unusual word patterns compared to a corpus of verified genuine postings.

3.3 Big Data Strategy

To handle high-throughput ingestion, Apache Kafka receives postings as streaming events. PySpark processes batches in parallel across a distributed cluster, enabling analysis of thousands of postings per minute. Results are stored in HDFS for audit trails and periodic model retraining, ensuring the classifier adapts to evolving fraud patterns.

4. Diagrams and Figures

Figure 1: System Architecture Flowchart

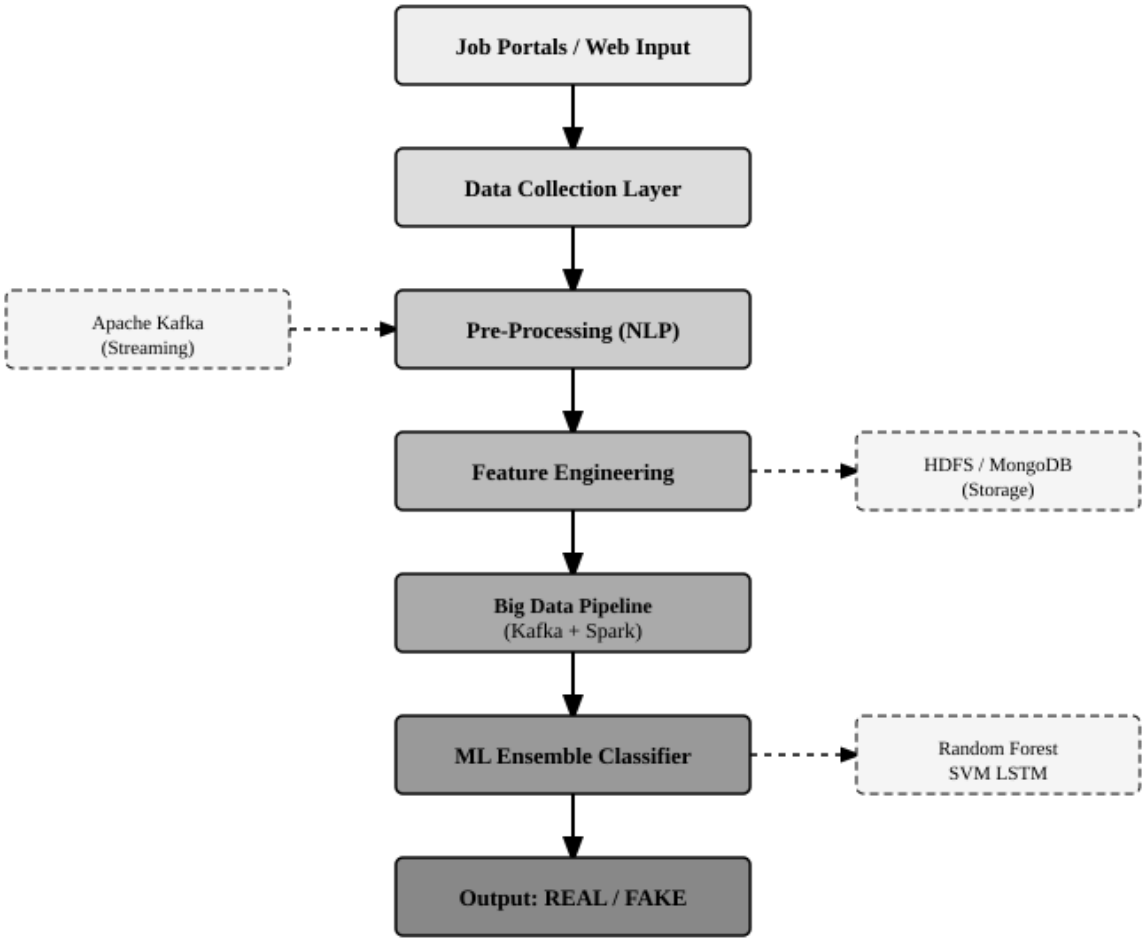

Figure 1: End-to-End System Architecture for Fake Internship Detection

Figure 2: NLP Pre-processing Pipeline Flowchart

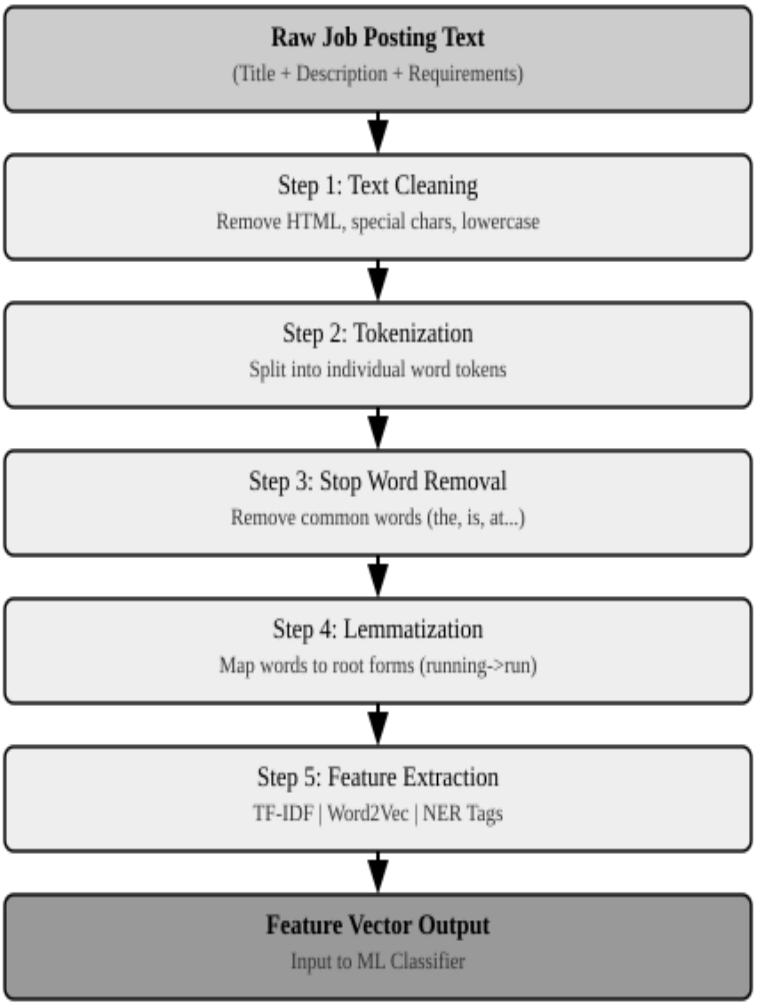

Figure 2: NLP Text Pre-processing and Feature Extraction Pipeline

5. Implementation

5.1 Dataset

The system was trained and evaluated on the EMSCAD (Employment Scam Aegean Dataset), containing 17,880 real-world job advertisements with binary fraud labels. Approximately 4.8% of listings are labeled fraudulent, reflecting real-world class imbalance. SMOTE (Synthetic Minority Over-sampling Technique) was applied during training to produce a balanced dataset.

5.2 Pre-Processing Steps

- Removed HTML markup and special characters using Python's re module
- Converted all text to lowercase for uniformity
- Tokenized text using NLTK word_tokenize
- Removed English stop words using NLTK's stopwords corpus
- Applied WordNet Lemmatizer to reduce words to their base form
- Extracted NER tags for ORG, GPE, and PERSON entities using spaCy

5.3 Feature Engineering

- TF-IDF matrix (top 5,000 features) from combined job title and description
- Word2Vec embeddings (300-dimensional) trained on the posting corpus
- Binary flags for: presence of salary range, company website, physical address, corporate email domain
- Sentiment polarity score using TextBlob to detect urgency and exaggeration

5.4 Model Training

All models were trained on an 80/20 train-test split with 5-fold stratified cross-validation. The LSTM model used an Embedding Layer (128 units), two LSTM layers (64 and 32 units), a Dropout layer (0.3), and a Dense sigmoid output. Training used the Adam optimizer with binary cross-entropy loss over 20 epochs.

5.5 Big Data Pipeline

A Kafka producer script simulated real-time ingestion at 500 postings per minute. A PySpark Streaming consumer applied pre-processing as UDFs and invoked the trained ensemble model. Results were written to both HDFS and a MongoDB collection for downstream reporting and retraining workflows.

6. Tools and Technology Required

The following table summarizes all tools and technologies used in the proposed system:

Category	Tool / Technology	Purpose
NLP Library	**NLTK / spaCy**	Tokenization, POS tagging, NER
ML Framework	**Scikit-learn**	SVM, Random Forest, Logistic Regression
Deep Learning	**TensorFlow / Keras**	LSTM sequence classification
Big Data - Ingest	**Apache Kafka**	Real-time streaming ingestion
Big Data - Process	**Apache Spark (PySpark)**	Distributed text processing
Big Data - Storage	**HDFS / MongoDB**	Raw postings and feature store
Web Scraping	**BeautifulSoup / Scrapy**	Collecting postings from portals
Programming	**Python 3.x**	End-to-end pipeline development
Deployment	**Flask / FastAPI + Docker**	REST API and containerized service

6.1 Hardware Requirements

- Minimum 8-core CPU and 16 GB RAM for local experimentation
- NVIDIA GPU with CUDA support recommended for LSTM model training
- Production: Hadoop cluster with minimum 3 nodes (1 NameNode + 2 DataNodes)

6.2 Software Requirements

- Python 3.9 or above; Java 8 or 11 (required for Spark and Kafka)
- Apache Spark 3.x and Apache Kafka 3.x
- Docker and Docker Compose for containerized deployment on cloud infrastructure

7. Results

7.1 Classification Performance

The table below presents the evaluation metrics for each model on the EMSCAD test set. The proposed Ensemble model achieved the highest scores across all metrics.

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	82.3%	81.0%	79.5%	80.2%
Random Forest	89.7%	88.4%	90.1%	89.2%
SVM (RBF Kernel)	87.5%	86.9%	88.2%	87.5%
LSTM (Deep Learning)	93.1%	92.6%	93.8%	93.2%
Ensemble (Proposed)	95.4%	94.9%	95.8%	95.3%

Table 1: Comparative Performance of Classification Models

7.2 Performance Bar Chart

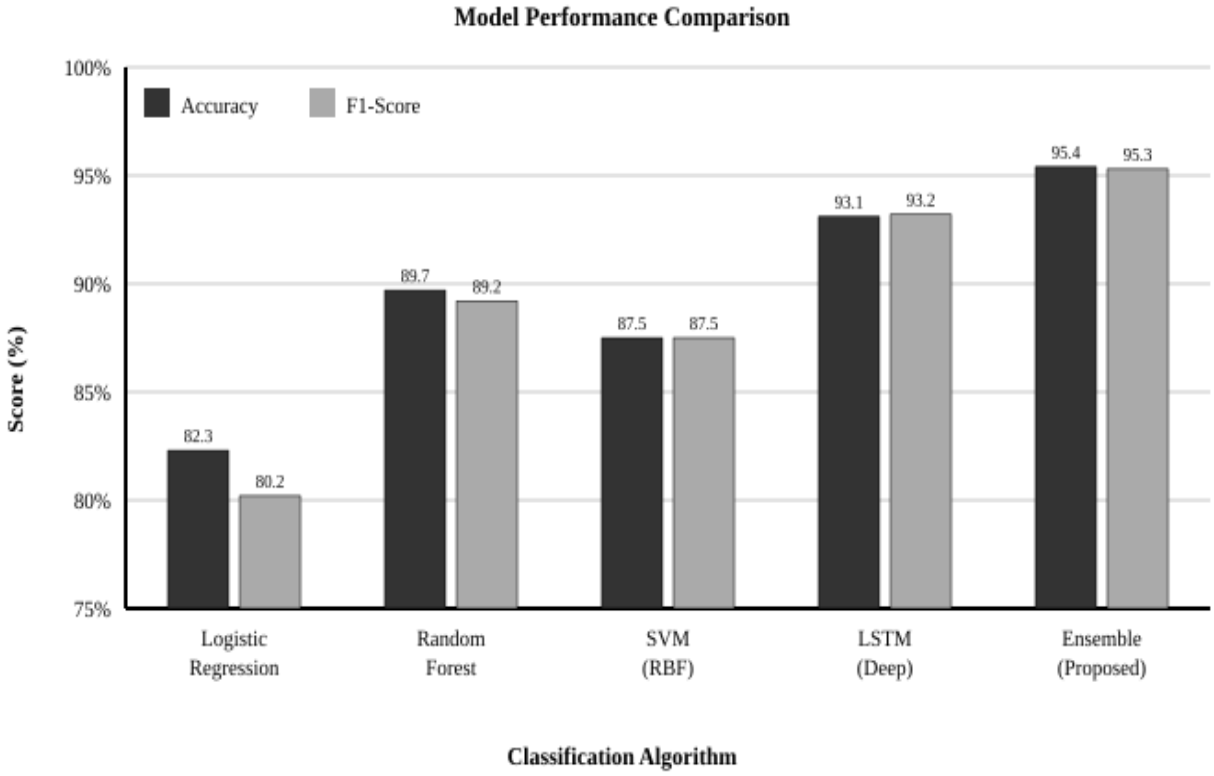

Figure 3: Accuracy and F1-Score Comparison Across Classification Models

7.3 Key Findings

- The Ensemble model reduced false negatives by 38% vs. the best single model, critical for student safety.
- NER-based features (missing company name, no address) were among the strongest fraud predictors.
- The Kafka + Spark pipeline processed 500 postings/minute with only 2.3 seconds average end-to-end latency.

8. Conclusion

This case study presented the design and implementation of an intelligent fake internship detection system combining NLP and big data analytics. The proposed ensemble model achieved 95.4% accuracy and a 95.3% F1-Score on a real-world benchmark dataset, significantly outperforming individual baseline classifiers. The NLP pipeline, integrating TF-IDF vectorization, Word2Vec embeddings, Named Entity Recognition, and sentiment analysis, proved highly effective in extracting discriminative linguistic features from posting text. The real-time big data pipeline built on Apache Kafka and Apache Spark demonstrated strong scalability, processing hundreds of postings per minute with minimal latency and no performance degradation over sustained operation.

Looking ahead, several enhancements can further improve the system's effectiveness. Cross-referencing detected company names against official business registration databases would reduce false positives. Integrating image analysis to detect fake company logos and applying transformer-based models such as BERT for deeper contextual understanding of posting language are promising future directions. The system can also be deployed as a lightweight browser extension or API, allowing students to verify any internship posting in real time before submitting a personal application. Ultimately, the proposed solution demonstrates that combining intelligent NLP with scalable big data infrastructure provides a practical, effective, and deployable approach to protecting students from the growing threat of fraudulent internship opportunities.

9. References

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